

GPU-Accelerated Agent-Based Simulation for Cell Dynamics

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*A Senior Thesis as a partial fulfillment of requirements
for the Bachelor of Science in Computer Science*

Department of Computer Science
Yale University
May 1, 2025

Acknowledgements

I am grateful to my advisor, Dr. Smita Krishnaswamy, for her guidance and encouragement throughout this project. Thanks to Xingzhi Sun and Jake Kovalic for introducing me to the agent-based modeling ideas that inspired part of this work.

Warm thanks to my friends in the Computer Science Department and my peers in Davenport College—our discussions over tea, coffee, or after a run were invaluable for troubleshooting and brainstorming.

Finally, I owe my deepest appreciation to my parents and brother for bearing with my sometimes obsessive focus on these simulations and for their constant support.

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Abstract

Life-like complexity often arises from the interplay of simple local rules, conserved resources, and adaptive control. In this thesis, we develop and analyze four grid-based experiments that progressively integrate these principles:

First, a *reaction–diffusion chemistry* model embeds virtual “cells” with minimal internal chemistry and gene regulatory–network proxies in a seven-species diffusive medium, yielding emergent chemotactic trails, tandem motility, and resource-limited survival strategies.

Second, a *GPU-accelerated physics cellular automaton* abstracts material phases (solid, liquid, gas, void) into per-pixel “solidity” parameters; simple gravity- and diffusion-like exchanges executed via OpenCL produce bubble nucleation, buoyant rise, and phase separation at a $\sim 55\times$ speedup over a NumPy reference.

Third, *GoLEM* augments Conway’s Game of Life with explicit energy–mass stores and an evolvable neural-CA policy: births consume, deaths release energy, and a learned routing network stabilizes novel still-life equilibria and transient oscillators under conservation constraints.

Finally, a *NEAT-driven grass–prey coevolution* simulates autotrophs and herbivores whose neural controllers evolve via neuroevolution, producing phases of directional grass expansion, prey infiltration, overexploitation, and predator-driven collapse.

Together, these models demonstrate that minimal, physically grounded rules—combined with local decision-making and evolutionary adaptation—can recreate core hallmarks of metabolism, movement, growth, and open-ended novelty in silico.

1 Introduction

Artificial life (ALife) seeks to understand how complex, lifelike behaviors can emerge from simple, decentralized rules and interactions. From Conway’s Game of Life [1] and Turing’s reaction–diffusion morphogenesis [2], we know that rich patterns—gliders, spots, stripes—can arise without central control. More recent work on neural cellular automata [3] and neuroevolution (e.g. NEAT [4]) has shown that embedding adaptivity and evolution into these substrates unlocks even greater behavioral complexity.

In this thesis, we build a unified sandbox of four experiments that progressively layer on conservation laws, internal control, and evolutionary dynamics:

1. **Reaction–Diffusion Chemistry with GRNs:** Virtual “cells” endowed with minimal internal chemistry and small gene regulatory–network proxies inhabit a seven-species reaction–diffusion medium. Cells sense gradients, move, metabolize, and divide, revealing emergent chemotactic trails and survival strategies.
2. **GPU-Accelerated Physics CA:** Inspired by the Biomaker CA [5], we abstract material phases (solid, liquid, gas, void) via a single “solidity” parameter per pixel. A simple OpenCL kernel executes gravity- and diffusion-like exchanges, yielding bubbles, buoyancy, and phase separation at 55× speedup over a CPU baseline.
3. **GoLEM—Game of Life with Energy–Mass:** We reinterpret Conway’s CA through energy–mass conservation and an evolving neural CA policy, where births consume energy and deaths release it. This produces novel still-lives and transient oscillators stabilized by resource routing.
4. **NEAT Coevolution of Grass & Prey:** A minimal ecosystem of autotrophs and herbivores evolves sensory-motor neural controllers (via NEAT) to forage, reproduce, and survive. Emergent phases of grass patching, prey infiltration, and eventual predation-driven collapse illustrate ecological arms races and failure modes.

Together, these four models demonstrate that simple rules—when coupled with conserved quantities, local decision-making, and evolutionary adaptation—can recreate core hallmarks of life: metabolism, movement, growth, and open-ended novelty. In the chapters that follow, we review relevant prior work, present our methods, analyze the results, and discuss how this synthetic-biology sandbox may inform both ALife theory and computational models of multicellular dynamics.

2 Background

2.1 Cellular Automata and Emergence

Cellular automata (CA) are discrete spatial models in which simple local rules applied to grid cells can produce complex global behavior. The most celebrated example is Conway’s Game of Life, a zero-player CA where each cell is either “alive” or “dead” and evolves based on its eight neighbors. Despite its minimal update rules, Life exhibits rich emergent phenomena—gliders, oscillators, and even universal computation—demonstrating that complexity can arise from purely local interactions [1].

Parallel to CA, Alan Turing introduced the concept of reaction–diffusion systems, in which continuous fields of chemical concentrations—morphogens—diffuse and react to form patterns such as spots and stripes in embryonic development [2]. Turing’s theory showed that spatial self-organization need not rely on preexisting templates, but can emerge from diffusion coupled with nonlinear reactions. Both CA and reaction–diffusion models laid foundational principles for understanding how simple physical or chemical rules can generate lifelike structure.

2.2 Continuous-State and Neural Cellular Automata

Subsequent work extended binary CAs into continuous-state and learned-update frameworks. Lenia is a two-dimensional CA with continuous space–time–state and generalized local rules, supporting a remarkable diversity of self-organizing patterns akin to virtual microscopic organisms [3]. Lenia’s fuzzy, resilient lifeforms reveal how analog values and kernel-based update functions can enrich the CA paradigm.

More recently, Growing Neural Cellular Automata employ small neural networks as the local update rule itself, learning to grow and regenerate target patterns by adjusting weights through gradient-based optimization [6]. These neural CAs blur the line between handcrafted and learned dynamics, showing that learned reaction–diffusion–style rules can produce robust self-organizing behaviors far beyond manually designed kernels.

2.3 Agent-Based Evolution and Neuroevolution

Another major strand in artificial life endows agents with internal controllers—often neural networks—that evolve under selection. Polyworld demonstrated a 2D digital ecosystem where agents with evolving neural brains learn behaviors such as pursuit, evasion, and cooperation solely through natural selection and resource competition [7]. More gener-

ally, neuroevolution techniques like NEAT (NeuroEvolution of Augmenting Topologies) evolve both network weights and topologies, enabling controllers to complexify incrementally and solve challenging tasks [4].

In three-dimensional virtual environments, Sims evolved creatures whose morphologies and neural control systems coadapt to compete for resources under realistic physics [8]. These agent-based approaches illustrate how open-ended evolution of behavior and form emerges when neural controllers are free to mutate and recombine, driven by environmental interactions rather than explicit fitness functions.

2.4 Parallelism and GPU Acceleration in Artificial Life

As artificial-life models grow in size and dimensionality, computational efficiency becomes critical. Falling-sand simulations and multi-phase cellular automata have long been used to mimic granular and fluid dynamics, but their CPU-bound implementations often limit scale and interactivity. The Biomaker CA framework introduced a richly typed plant-body CA, yet remained constrained by sequential updates on the CPU [5].

Leveraging GPU parallelism via OpenCL or CUDA allows data-parallel cellular updates across millions of cells each timestep, enabling real-time exploration of large CA worlds. GPU-accelerated CAs have been applied to fluid simulations, reaction–diffusion, and agent-based models, offering orders-of-magnitude speedups over NumPy baselines. This combination of physical realism, evolutionary control, and hardware acceleration provides a powerful platform for sandboxing emergent life-like phenomena.

3 Methodology

3.1 Reaction–Diffusion Chemistry with Virtual Cells & GRNs

Building on agent-based models of cancer and immune dynamics (e.g. the Krishnaswamy Lab’s `EuclideanSimulation`), this experiment abstracts away biochemical and cell-type specifics to focus on fundamental life principles. The two-dimensional grid carries seven continuous species—water (H_2O), carbon dioxide (CO_2), oxygen (O_2), sugar, ATP, protein, and a generic mitogen—that diffuse per Fick’s law and react via a unified reaction list with both background (uncatalyzed) and cell-catalyzed rates.

Each cell occupies one grid site and maintains **three functional channels** for every chemical:

1. **Reaction channel (enzymes):** positive values catalyze production reactions; negative values catalyze consumption.
2. **Transport channel (transporters):** controls active import or export against local gradients.
3. **Motor channel (motors):** dictates energy-driven movement toward or away from chemical cues.

Channels are driven by a small, evolvable neural network that takes environmental concentrations and a cell’s current channels as inputs and outputs channel update values at an ATP cost—abstracting gene-to-protein translation and receptor dynamics into a unified, energy-conscious decision process. When mitogen and resource thresholds are met, cells divide—splitting chemical resources and mutating network weights—while ATP depletion or lifespan limits trigger death and release of internal contents back into the field.

This minimal yet biologically inspired framework emphasizes *metabolism*, *chemotaxis*, *transport*, and *division* as the core capabilities of life, enabling emergent gradient formation, differentiation-like cycles, and synthetic-biology behaviors without hard-coding specialized cell types.

3.2 GPU-Accelerated Physics Simulation

This experiment takes inspiration from the Biomaker CA framework [5], which simulates a richly typed biome via simple, local cellular-automaton rules. Rather than hard-coding several cell types (stem, leaf, root, seed, air, dirt, etc.), we abstract every material down

to its *intermolecular force* characteristics. Each pixel’s RGBA color represents a four-channel “phase” vector (solid, gas, liquid, void), and we assign each phase a **solidity** parameter that proxies bond strength—strong ionic/covalent bonds for solids, hydrogen bonds for liquids, and weak London forces for gases.

At each time step, an OpenCL kernel applies two core local exchanges:

1. **Gravity-like exchange** with the pixel above/below, clamped by the neighbor’s solidity. This captures how denser or more strongly bonded materials resist buoyant motion.
2. **Diffusive exchange** with the four orthogonal neighbors, scaled by the inverse of solidity. This encodes how weakly bonded phases (e.g. air) spread rapidly, while strongly bonded phases (e.g. stone) remain localized.

By tuning solidity values (e.g. 0.999 for rock, 0.5 for water, 0.001 for air), the system self-organizes into lifelike material interactions—liquids pool, solids pile, and bubbles form as low-solidity “air” rises through the higher-solidity “water” matrix. A NumPy reference implementation on the CPU highlights the dramatic throughput advantage of GPU compute (55× speedup).

This experiment unifies falling-sand dynamics and fluid behavior, which could be useful in future artificial-life models that integrate material physics with metabolism, chemotaxis, and multicellular mechanics.

3.3 GoLEM: Game Of Life with Energy–Mass Equivalence

This experiment reimagines Conway’s Game of Life as a miniature physics sandbox governed by conservation laws and entropy. Inspired by Einstein’s $E = mc^2$, we treat mass and energy as interconvertible yet globally conserved quantities: a cell birth consumes one unit of energy, and a cell death releases that energy back into the field. In the absence of life, the system tends toward an entropy-increasing equilibrium under standard Life rules.

To capture how living patterns locally decrease entropy, each cell carries both mass and energy stores and participates in a learned, decentralized energy-routing process. We augment the binary GoL grid with a continuous energy channel plus a set of hidden “genome” channels. A neural-policy CA—analogous to Growing Neural CAs [3]—reads local mass, energy, and hidden channels, then directs energy flows across each 3×3 neighborhood, conserving the total and modeling how organisms “hijack” physics to persist ordered patterns. Cells inherit their parents’ crossed-over policy parameters upon reproduction and lose them at death, enabling evolutionary adaptation of energy-management strategies.

By fusing Life’s simple birth/death topology with energy–mass conversion, entropy considerations, and an evolvable neural CA policy, GoLEM aims to exhibit richer emergent phenomena—sustained oscillators, self-organized energy waves, and resource-driven

pattern formation—that mirror how real organisms defy equilibrium to survive and replicate under thermodynamic constraints.

3.4 NEAT Coevolution of Grass and Prey

In this experiment, we simulate an abstract ecosystem of autotrophic “grass” and herbivorous “prey” on a toroidal grid, with both species’ behaviors governed by evolvable neural controllers optimized via the NEAT algorithm [4]. Each organism carries a genome that encodes a Compositional Pattern-Producing Network (CPPN), which in turn generates a phenotype neural network mapping local sensory inputs to action outputs.

Sensory Inputs

- **Grass:** samples a 3×3 RGB patch of its immediate neighborhood to decide propagation and energy expenditure.
- **Prey:** senses a 9×9 RGB field, enabling foraging and threat avoidance over a wider area.

Actions

- **Movement:** networks output $\Delta x, \Delta y$ vectors; prey use these to move toward nutrients or away from threats, consuming energy proportional to distance moved.
- **Asexual reproduction:** when energy exceeds a threshold, an organism clones itself into an adjacent empty cell.
- **Sexual reproduction (Prey only):** if two prey mutually point at each other (each network’s $\Delta x, \Delta y$ directives align) and both have sufficient energy, they produce a shared offspring in a neighboring empty site.

Energy Dynamics

- Grass regenerates energy passively and must maintain a minimum to reproduce.
- Prey gain energy by consuming grass and lose energy through movement and reproduction. Death occurs when energy falls to zero.

By coevolving these simple sensory-motor networks, grasses develop directional growth patterns to reduce predation risk, while prey evolve tactics—clustering, dispersal, ambush—and mutual breeding behaviors. Over thousands of generations, this setup yields rich ecological dynamics: spatial patchiness, herding, and arms-race feedback loops, illustrating how minimal neural controllers under selection can generate complex, lifelike interactions.

4 Results

4.1 Reaction–Diffusion Chemistry with Virtual Cells & GRNs

Environmental Metrics

Timestep	H ₂ O	CO ₂	O ₂	Sugar	ATP	Protein	Mitogen
0	1.00000	1.00000	1.00000	1.00006	1.99998	0.000001	0.06094
2999	0.98448	0.98409	0.98109	0.98825	1.82319	0.02165	0.00383

Table 4.1: Mean concentrations at initial and final timesteps.

Population Metrics (at $T = 2999$)

- Live-cell count: 2
- Mean internal ATP: 0.9116

Qualitative Observations

1. **Rapid mitogen-driven expansion:** Cells adjacent to the central mitogen patch divide explosively (Figure 4.1a).
2. **Trail-forming motility and tandem movement:** Around $T \approx 860$, motile cells travel in coordinated pairs leaving chemical trails (Figure 4.1b).
3. **Long-term survivors:** By $T = 2990$, only two cells remain (Figure 4.1c), suggesting evolved survival strategies.

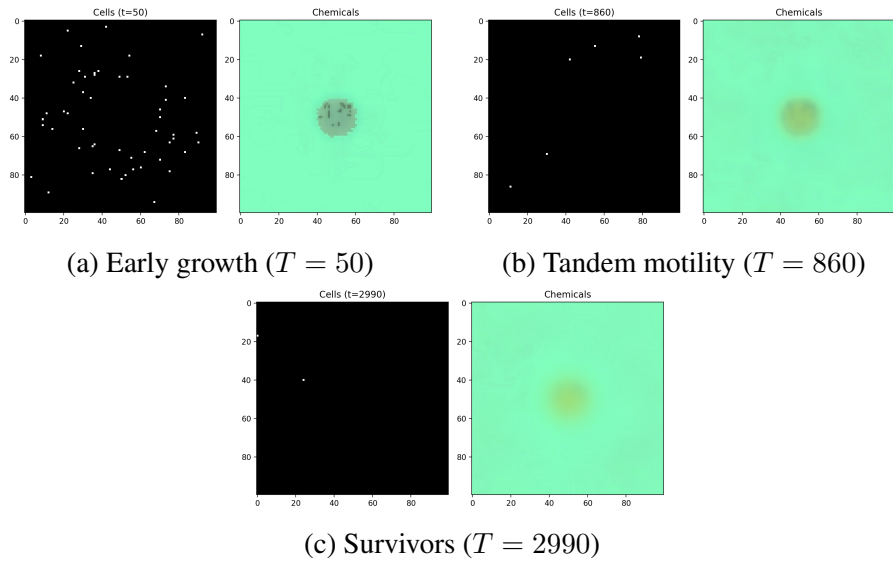


Figure 4.1: Snapshots of the reaction-diffusion CA at key timesteps.

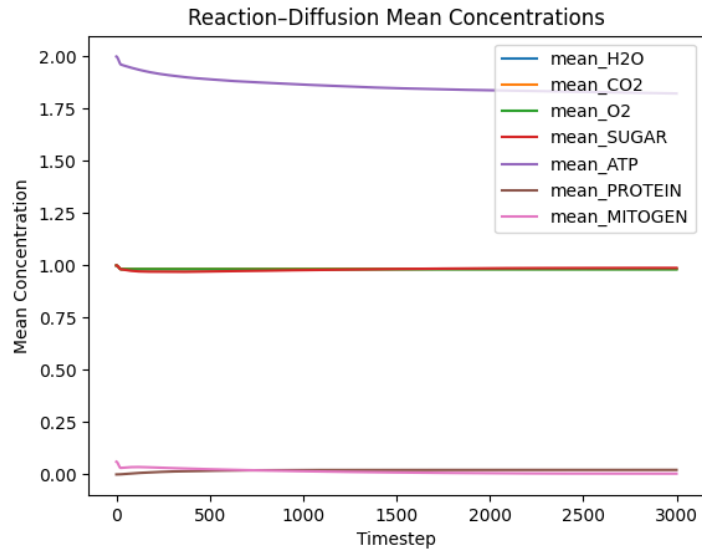


Figure 4.2: Mean chemical concentrations over time.

4.2 GPU-Accelerated Physics Simulation

Performance Metrics

Metric	Value
Avg. GPU time/step	1.36 ms
Avg. CPU time/step	74.51 ms
GPU/CPU Speedup	$\times 54.8$

Table 4.2: Per-step timing comparison.

Qualitative Observations

1. **Bubble-like air pockets:** Low-solidity “air” pockets coalesce and rise through liquid and solid phases (Figure 4.3a).
2. **Persistent liquid trapping:** High-solidity solids confine liquids beneath them (Figure 4.3b).
3. **Layered material interactions:** Solids settle, liquids pool, and gases dissipate over time (Figure 4.3c).

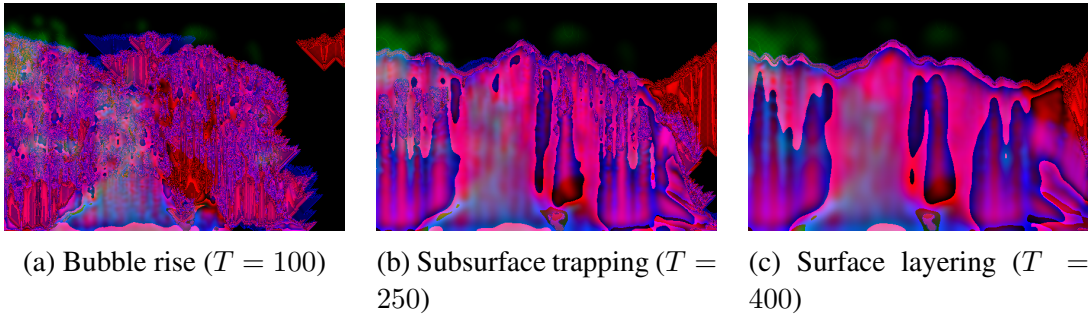


Figure 4.3: Physics CA at selected timesteps.

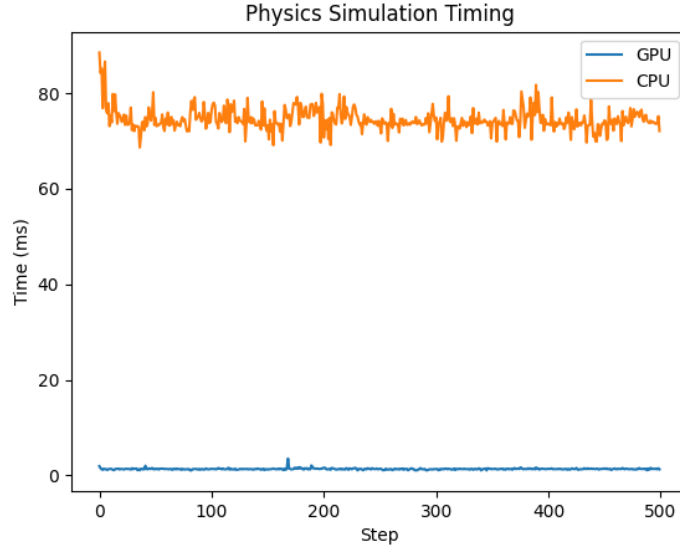


Figure 4.4: GPU vs. CPU per-step timing.

4.3 GoLEM: Game Of Life with Energy–Mass Dynamics

Equilibrium Metrics

Timestep	Mass	Energy	Conserved	Live cells
1	3201	1349	4550.0	3201
10	2485	2065	4549.9	2485
60	2466	2084	4550.0	2466
100	2466	2084	4550.0	2466

Table 4.3: Mass, energy, and live-cell counts at key steps.

Key Observations

1. **Glider-like oscillators:** Mobile structures traverse the grid (Figure 4.5a).
2. **Still-life equilibrium:** By $T \approx 60$, gliders burn out, leaving novel still-lives (Figure 4.5b).
3. **Conserved dynamics:** After $T = 60$, mass and energy remain nearly constant.

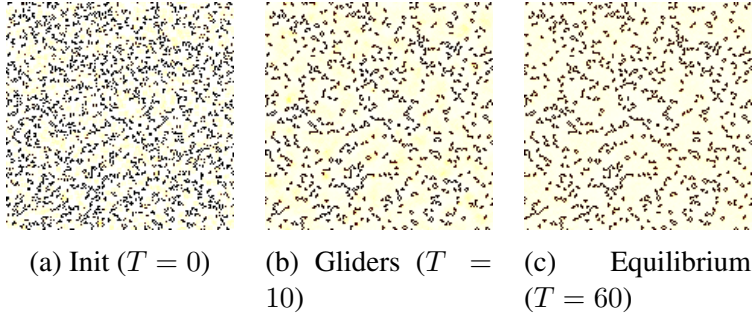


Figure 4.5: GoLEM snapshots at representative times.

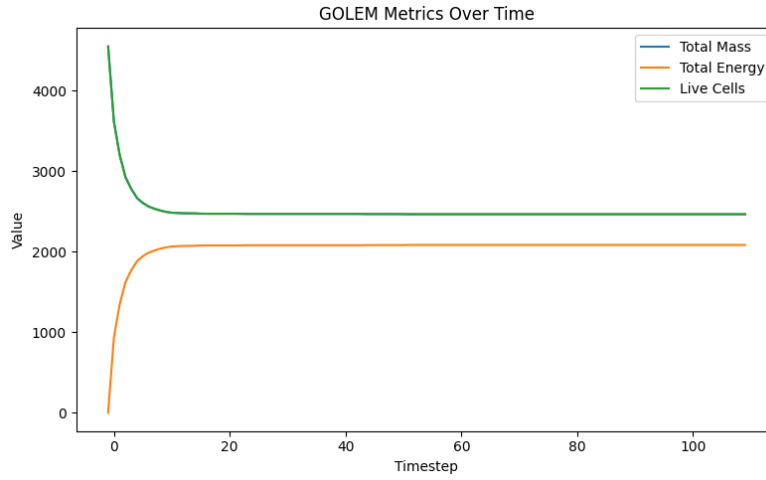


Figure 4.6: Total mass, energy, and live cells over time.

4.4 NEAT Coevolution of Grass & Prey

Population Metrics

Timestep	Grass	Prey
0	20	100
800	737	152
2000	412	215
2325	150	0
4999	180	0

Table 4.4: Grass and prey populations over time.

Qualitative Observations

1. **Directional grass expansion:** Grass evolves elongated patches (Fig. 4.7a).

2. **Prey infiltration:** Prey cluster and then penetrate grass edges (Fig. 4.7b).
3. **Infestation outbreak:** Dense prey proliferation overruns grass (Fig. 4.7c).
4. **Prey extinction:** Resource exhaustion leads to die-off (Fig. 4.7d).

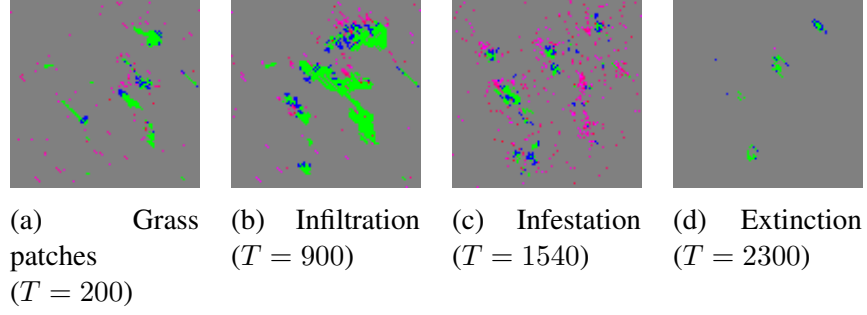


Figure 4.7: Grass-prey dynamics at key timesteps.

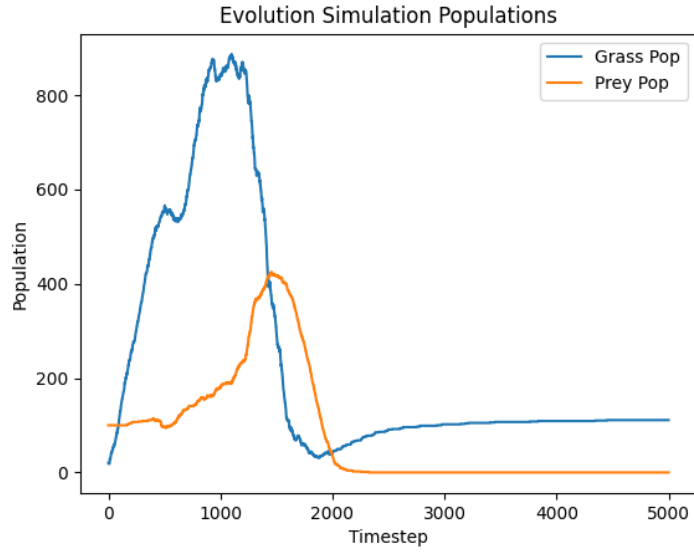


Figure 4.8: Grass and prey populations over time.

4.5 Discussion

Across these four experiments, we observe how simple local rules, when combined with conserved quantities and evolvable control, give rise to rich, lifelike behaviors:

1. Emergence through Conservation Laws

- In both the Reaction–Diffusion Chemistry and GoLEM experiments, enforcing conservation of mass and energy introduces nontrivial steady states.
- The ChemRD model collapses without sustained inputs, whereas GoLEM’s routing policy stabilizes populations into persistent still-lives, illustrating how active resource management alters CA dynamics.

2. Chemotactic and Motility Behaviors

- Cells in the Chem RD model exhibit trail-following and tandem movements purely through transporter and motor channels driven by GRN outputs—no explicit movement code was needed.
- In GoLEM, glider-like oscillators metamorphose into novel energy-bounded stills, indicating that neural routing of conserved resources can stabilize or extinguish classical CA motifs.

3. Phase Separation and Material Self-Organization

- The GPU physics CA, inspired by Biomaker CA, demonstrates how a single “solidity” parameter per phase yields realistic layering, bubble nucleation, and buoyant rise.
- A $\sim 55\times$ GPU speedup over CPU makes real-time exploration feasible, highlighting the importance of hardware acceleration for large-scale artificial-life simulations.

4. Coevolutionary Dynamics and Failure Modes

- Grass and prey coevolution under NEAT pressures yields sequential strategies—directional expansion, prey infiltration, infestation—mirroring ecological arms races.
- The prey-driven extinction event underscores the need for regulatory feedback (e.g., resource signaling, density-dependent costs) to sustain long-term coexistence.

5. Interplay of Biology and Abstraction

- Abstracting biological processes (e.g., metabolism, transport, division) into minimal channels and networks allows focus on fundamental principles—conservation, diffusion, evolution—without overfitting to specific biological systems.

- Yet, each model remains extensible: new reactions, feedback loops, or physical forces can be added to probe deeper into the origins of life-like phenomena.

Overall, these experiments chart a spectrum from purely chemical pattern-formation to agent-based coevolution, unified by core ideas of resource conservation, local decision-making, and emergent organization. They demonstrate that even vastly simplified proxies for metabolism, physics, and evolution can produce behaviors reminiscent of real life, laying a groundwork for future artificial-life frameworks that integrate chemistry, mechanics, and adaptive control.

5 Related Work

5.1 Reaction–Diffusion and Synthetic Embryogeny

Early reaction–diffusion models trace back to Turing’s landmark work, demonstrating how interacting morphogens can spontaneously form spatial patterns in homogeneous media [2]. The Gray–Scott variant and its kin have since been used to generate spots, stripes, and other morphogenetic motifs. In artificial life, the integration of reaction–diffusion with cellular agents has been explored to more closely mimic biological cells. For example, Dembélé et al. combined gene regulatory networks (GRNs) with spatial proliferation rules to allow cells to self-adapt metabolic behaviors in response to nutrient fields [9]. However, many of these systems treat cell catalysts and transport mechanisms in a fixed, hand-coded manner.

Our contribution: We abstract the full set of internal and external chemical species into unified reaction lists and evolvable neural controllers, enabling cells to regulate metabolism, transport, and movement through the same minimal GRN framework without bespoke reaction modules.

5.2 Neural Cellular Automata and Evolving Update Rules

Continuous-state CAs such as Lenia [3] extend binary Life by using analog values and convolutional kernels to support a rich taxonomy of “virtual lifeforms.” Growing Neural Cellular Automata further push this boundary by replacing handcrafted kernels with small neural networks, trained to regenerate target patterns or images [6]. In parallel, open-ended CA evolution—exemplified by Sayama’s Evoloops [10]—demonstrates that self-reproducing and evolving structures can arise within discretized automata. Yet, most neural CA research focuses on pattern synthesis rather than genuine birth-death dynamics or resource constraints.

Our contribution: GoLEM integrates an evolvable neural CA policy with mass–energy conservation laws, marrying the continuous update philosophy of neural CAs with true reproductive and energetic transactions, thus stabilizing novel still-life equilibria.

5.3 GPU-Accelerated Material Cellular Automata

Falling-sand simulations and multi-phase CAs have provided compelling visuals of granular and fluid behavior, but their CPU-bound implementations limit resolution and interactivity. The Biomaker CA framework [5] implements several plant-body types within a

richly typed CA but remains constrained by sequential updates. GPU implementations of fluid and sand CAs (e.g. via OpenCL or CUDA) achieve significant speedups but often rely on separate rule sets for each material phase.

Our contribution: We unify all material phases under a single “solidity” parameter and apply two general local exchange laws—gravity-like and diffusion-like—across four RGBA channels. This abstraction, executed in parallel on the GPU, reproduces bubble nucleation, buoyant rise, and phase separation at real-time frame rates, highlighting a minimalist yet effective approach to material physics in ALife.

5.4 Coevolutionary Agent Ecosystems

Agent-based evolution has a rich history: Tierra [11] created a primordial soup of self-replicating programs, Polyworld endowed agents with evolving neural controllers to explore emergent behaviors [7], and Sims evolved 3D virtual creatures with coadapted morphology and control [8]. NEAT [4] introduced a powerful methodology to evolve both topology and weights of neural networks, enabling complex behaviors without predefined architectures. These systems demonstrate open-ended novelty but often require custom fitness functions or are limited to single-species dynamics.

Our contribution: The NEAT Grass–Prey simulation constructs a two-species co-evolutionary model on a toroidal grid, with both asexual and mutual sexual reproduction rules. By evolving sensory-motor CPPNs via NEAT in a resource-limited environment, our model produces sequential ecological strategies and highlights failure modes like self-extinction, emphasizing the need for regulatory feedback for sustainable ecosystems.

6 Conclusions

This thesis explored four distinct yet thematically linked experiments that demonstrate how simple local rules, conserved resources, and evolvable control can give rise to rich, lifelike phenomena:

1. **Reaction–Diffusion Chemistry with Virtual Cells & GRNs** showed that embedding cells with minimal internal chemistry and gene regulatory networks into a reaction–diffusion medium yields emergent motility, chemotactic trails, and resource–limited population dynamics. While the system collapsed without sustained nutrient inputs, it evidenced genuine gradient–driven behavior and hinted at strategies for indefinite persistence.
2. **GPU–Accelerated Physics Simulation** distilled material interactions to a single “solidity” proxy per phase and two local exchange laws, reproducing bubble formation, buoyant rise, and phase separation. A $\sim 55\times$ GPU speedup over CPU illustrates the power of data–parallel compute for scaling artificial–life environments.
3. **GoLEM: Game of Life with Energy–Mass Dynamics** reinterpreted Conway’s CA through the lens of energy–mass equivalence and conservation. By coupling birth/death rules with an evolved neural routing policy, GoLEM transitions from transient glider oscillations to novel still–life equilibria, revealing how resource constraints stabilize patterns beyond classical GoL.
4. **NEAT Coevolution of Grass & Prey** implemented a minimalist ecosystem in which grasses and herbivores coevolve sensory–motor networks under NEAT. The run of sequential strategic phases—directional grass patches, prey infiltration, infestation, and eventual prey extinction—highlights both the creative potential and pitfalls of open–ended evolution without regulatory feedback.

Taken together, these studies underscore that even highly abstracted models—whether chemical, physical, or ecological—can reproduce core aspects of living systems: metabolism, movement, growth, adaptation, and self–organization. Conserved quantities (mass, energy) and simple control architectures (GRNs, neural policies, NEAT–evolved networks) provide a unifying framework for artificial–life research. This work lays a foundation for future investigations into sustained self–organization, multi–scale coupling of chemistry and mechanics, and evolvable life–like systems that bridge the gap between minimal models and biological complexity.

7 Future Work

Building on these four experiments, several avenues can deepen our understanding of artificial-life systems and push the models closer to biological realism:

1. Dynamic Resource Influx & Homeostasis

- Introduce periodic or spatially distributed nutrient and mitogen sources in the Reaction–Diffusion Chemistry model to sustain populations and study steady–state diversity.
- Implement feedback-regulated background reactions (e.g., photosynthetic sugar synthesis) to model external energy inputs.

2. Parameter Sweeps and Phase Diagrams

- Systematically vary key parameters (diffusion rates, reaction cost coefficients, energy leak rates, solidity values) to map emergent behavior regimes (e.g., pattern formation, oscillatory phases, extinction thresholds).
- Employ automated grid searches and sensitivity analysis to identify bifurcation points and critical transitions in each model.

3. Enhanced GRN Architectures

- Expand the gene regulatory network model with additional channel types (e.g., signaling peptides, receptor analogs) to explore richer cell–cell communication and differentiation protocols.
- Compare discrete logical GRNs versus continuous neural-net proxies to assess trade-offs in evolvability and behavioral complexity.

4. 3D Extension and Multiscale Coupling

- Generalize the Reaction–Diffusion and Physics CAs to three dimensions, coupling volumetric diffusion with 3D cell populations or fluid–solid interactions.
- Integrate biomechanical forces (e.g., adhesion, pressure) with chemical and energetic dynamics to model tissue–like growth and morphogenesis.

5. Stabilizing Coevolutionary Feedbacks

- In the NEAT ecosystem, introduce negative feedback mechanisms (e.g., resource signaling, disease spread, density–dependent reproduction costs) to prevent runaway extinction and achieve long–term coexistence.

- Investigate tournament and island–model coevolution setups to foster spatially heterogeneous evolutionary dynamics.

6. Benchmarking & Real–Time Interaction

- Package the GPU–Accelerated Physics simulation as an interactive tool (e.g., with UI sliders for solidity or diffusion) to explore material behaviors in real time.
- Profile and optimize neural routing CA implementations for deployment on WebGL or mobile GPUs, enabling broader accessibility.

7. Integration into a Unified Artificial–Life Framework

- Develop a modular platform where chemical, physical, and ecological CAs can be composed—e.g., embedding motile cells from the reaction–diffusion model into GoLEM or coupling grass–prey agents with underlying fluid flows.
- Leverage standardized interfaces (e.g., JSON or YAML configs) to orchestrate hybrid simulations and facilitate reproducible research.

By pursuing these directions, we can progressively bridge minimal artificial-life models and more comprehensive *in silico* ecosystems, offering insights into the fundamental principles that underlie living systems and their emergence from simple rules.

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